UDAIPUR SMART CITIES & TECHNO INDIA NJR PILOT | PHASE 1 ENGINEERING SPECIFICATION | AAROGYAJAL PLATFORM INTEGRATION

Fateh Sagar Lake Smart Water & Health Surveillance System

Integrated Sensor Selection, Hardware Architecture, Disease Detection & Telemetry Blueprint

LEAD AUTHORS	TARGET SITE	AFFILIATION	DEPLOYMENT DATE
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6 Probes	10 Minutes	48 Hours	4G LTE
Core Continuous Array	Telemetry Ingestion Rate	Solar Battery Autonomy	Encrypted TLS Ingestion

1. Purpose, Scope & Resolution of Deployment Parameters

This document integrates the *Phase 1 Sensor Selection Report* authored by Nawdeep with the complete *Aarogya Jal Telemetry, ML & Public Health Architecture*. It resolves all open scope parameters for deployment at **Fateh Sagar Lake, Udaipur** under the Techno India NJR / Udaipur Smart Cities framework.

Target Deployment Site — **Udaipur, RJ**

Location: Fateh Sagar Lake (Madar canal inflow feeding mouth & Public Health Engineering Department [PHED] drinking water intake jetty).
Dual Purpose: Protection of municipal drinking water supply intake for Udaipur city + environmental lake ecology and algal bloom tracking.

Regulatory Standards Baseline — **CPCB & BIS**

Lake Baseline: CPCB Designated Best Use *Class C* (Drinking water source after conventional treatment & disinfection) and *Class B* (Outdoor bathing).
Intake Manifold: Cross-referenced against *BIS 10500:2012* drinking water thresholds.

2. Phase 1 Core Sensor Suite: Lake Health & Municipal Intake Array

Six primary sensors cover the limnological and public health indicators of Fateh Sagar. As established in the engineering review, hobby-grade probes are excluded for pH, conductivity, and dissolved oxygen to guarantee field longevity:

#	SENSOR	READS	LIMNOLOGICAL & MUNICIPAL SIGNIFICANCE	TARGET SPECIFICATION & GRADE
1	**Dissolved Oxygen (DO)**	mg/L or % saturation	Single best indicator of lake health: flags organic sewage inflow, algae collapse, and oxygen starvation before municipal intake.	Optical luminescent probe preferred; ±0.2 mg/L target accuracy.
2	**pH Sensor**	pH units (0–14)	Tracks diurnal swings driven by algal photosynthesis (daylight consumption of CO_2) vs. acidic sewage/industrial discharge.	Industrial glass electrode; target ±0.05 pH (competition spec).
3	**Turbidity Module**	NTU (0–3000)	Measures suspended silt and runoff entering from Aravalli hills after rainfall; detects pathogen carrier vehicles.	Infrared 90° nephelometric scatter; calibrated in NTU, not 0–100%.
4	**Conductivity & TDS**	µS/cm, mg/L TDS	Measures dissolved salts; distinguishes pure rainwater dilution from untreated domestic sewage ingress.	Range to 5000 µS/cm; target ±20 mg/L TDS (ICT Grand Challenge spec).
5	**Water Temperature**	°C (-55 to +125)	Drives bacterial incubation kinetics; mandatory thermal compensation input for pH, DO, and conductivity probes.	DS18B20 digital stainless steel probe; ±0.5°C accuracy.
6	**Water Level**	cm or meters	Tracks lake water balance, storm inflow surges, evaporation rates, and sluice gate discharge to Swaroop Sagar.	Submerged hydrostatic pressure transducer or rigid ultrasonic mount.

3. Detailed Sensor Physics & Operational Use Cases in Fateh Sagar

Each sensor addresses a specific limnological mechanism within Fateh Sagar Lake. Machine learning models require all six concurrent streams to disentangle normal diurnal biological cycles from acute contamination events:

1. Dissolved Oxygen (DO) — Optical Luminescence

Use Case: In Fateh Sagar, healthy lake water exhibits an oxygen curve that rises during daylight (algal photosynthesis) and drops at night (respiration). A flat or collapsing DO curve is an immediate fingerprint of high organic sewage ingress, where decomposing microbes rapidly deplete oxygen.
ML Utilization: Forecasts localized oxygen collapse 48–72 hours ahead; serves as primary input for real-time Biochemical Oxygen Demand (BOD) soft-sensor estimation.
Operational Constraints: Optical luminescent probes drift significantly less than galvanic membrane probes and do not consume oxygen during measurement. Readings must be compensated for water temperature and salinity.

2. pH Dynamics — Industrial Glass Electrode

Use Case: Algal blooms in Fateh Sagar consume dissolved carbonic acid during daylight, driving pH up to 8.5–9.2. Conversely, domestic wastewater and septic inflows depress pH below 6.8.
ML Utilization: Learns the lake's seasonal diurnal baseline swing. Deviations outside the normal diurnal envelope indicate severe chemical or septic dumping.
Operational Constraints: Continuous immersion in bio-active water causes glass bulb scaling; requires automated threshold drift filtering and fortnightly buffer calibration.

3. Turbidity — Infrared Nephelometry

Use Case: Silt from the Aravalli catchment and stormwater runoff washes into Fateh Sagar via the Madar feeding canal. Pathogens (E. coli, Cryptosporidium) adhere to suspended colloidal silt.
ML Utilization: Correlates rainfall volume with turbidity surge lags; feeds organic decay estimates.
Operational Constraints: Biofilm accumulation on optical windows mimics a slow artificial rise in turbidity. Addressed via software baseline drift subtraction and optical window wipers.

4. Conductivity & Total Dissolved Solids (TDS) — AC Excitation Probe

Use Case: Distinguishes heavy rainwater inflows (which dilute lake conductivity) from urban sewage discharges (which cause sharp conductivity surges due to sodium, chloride, and dissolved electrolytes).
ML Utilization: Separates natural hydrological dilution from raw sewer breach events.
Operational Constraints: AC excitation pulses must be used instead of DC to prevent electrode polarization. TDS is derived as: TDS (mg/L) = k × EC (µS/cm), with k calibrated to Fateh Sagar water.

5. Water Temperature — DS18B20 Digital

Controls dissolved gas solubility (cold water holds more DO). Acts as the mathematical thermal compensation baseline for pH, DO, and conductivity probes. Detects deep-water thermal stratification.

6. Water Level — Hydrostatic Transducer

Tracks lake storage volume and inflow surges. Replaces the 500 cm limit of ultrasonic sensors with a submerged 0–10m gauge pressure sensor unaffected by surface wave action or wind noise.

4. The 4-Tier Disease Detection Architecture

A central challenge in water health surveillance is that **sensors cannot directly image or count pathogenic bacteria in real time**. Aarogya Jal resolves this through an integrated four-tier surveillance engine:

1. **Physical Proxies**
   24/7 telemetry detects the physical-chemical shocks that accompany fecal sewage intrusion: Turbidity ↑, DO ↓, Conductivity ↑.
2. **ML Risk Engine**
   Ensemble anomaly models calculate Pathogen Ingress Probability and Outbreak Risk Index (ORI) within 10 minutes.
3. **Biological Ground-Truth**
   Automated alert prompts on-demand rapid E. coli / Coliform enzymatic test strip (Aquagenx CBT) for direct proof.
4. **Municipal Alert Loop**
   Alerts PHED intake engineers to increase chlorination or halt intake pumps 48–72h before clinical cases emerge.

Specific Disease Diagnostic Matrix

DISEASE TARGET	PRIMARY PATHOGEN	REAL-TIME IN-SITU PHYSICAL FINGERPRINT	CONFIRMATORY MODALITY & MUNICIPAL ACTION
Cholera	*Vibrio cholerae*	Sharp DO collapse (rapid microbial aerobic respiration) + Turbidity surge + warm water (>24°C).	On-demand rapid coliform strip; immediate pre-chlorination dosing alert to PHED water treatment plant.
Diarrheal Outbreak	*E. coli*, Rotavirus, Cryptosporidium	Severe post-rain turbidity spike (>15 NTU) indicating silt runoff containing livestock/human waste.	Enzymatic β-glucuronidase test strip; coagulant dosage increase alert at water treatment filters.
Typhoid	*Salmonella typhi*	Elevated conductivity surge (raw domestic sewage line breach entering Madar canal) + depressed DO.	Lab sample verification; automated isolation and sluice diversion notice away from intake mouth.
Hepatitis A	Hepatitis A virus (HAV)	Chronic sewage mixing indicators: elevated baseline TDS, abnormal pH drift, low baseline oxygen.	Monthly viral PCR / coliform surrogate log; sanitary survey dispatched to lakefront commercial properties.
Algal Toxicity	Cyanobacteria (Microcystins)	Daytime pH spikes (>8.8) paired with high water temp, followed by acute nocturnal DO drops.	Microscopic bloom check; activated carbon dosing recommendation to eliminate cyanotoxins.

The Proactive 48-Hour Early-Warning Advantage

Traditional disease surveillance is **reactive**: authorities discover contamination only when hospitals report acute diarrheal admissions (3–5 days after exposure). In Fateh Sagar, Aarogya Jal flags fecal sewage intrusion at the lake station within **10 minutes of arrival**, allowing PHED treatment engineers to neutralize pathogens before contaminated water ever reaches the city tap distribution grid.

5. Sensor Technology Trade-Offs & Real-Time Soft-Sensor Surrogates

The sensor selection reflects an explicit compromise between measurement precision, equipment capital expenditure, and rural/municipal maintenance feasibility:

ISE (Ion-Selective) — Drift-Prone

Evaluation: Inexpensive for Nitrate/Ammonium, but suffers heavy calibration drift and ionic cross-interference. Requires frequent recalibration against known chemical buffers, making unattended lake deployment unreliable.

Wet-Chemical Units — Reagent-Heavy

Evaluation: Delivers lab-grade accuracy for Phosphate and Nitrate, but consumes liquid chemical reagents. Requires trained technicians for frequent refills and pump maintenance; fails unattended deployment criteria.

Optical / UV Probes — High CapEx

Evaluation: Accurate and reagent-free (e.g. s::can spectro::lyser), but costs ₹15 Lakh–₹35 Lakh per probe and requires high-pressure air compressors. Infeasible for widespread municipal lake stations.

Real-Time Soft-Sensor Surrogate: Biochemical Oxygen Demand (BOD)

As Nawdeep's report correctly emphasized, **true BOD cannot be measured by any real-time in-situ sensor** because standardized BOD_5 requires a 5-day dark biological incubation period. Aarogya Jal resolves this by deploying an **ML-Driven Soft Sensor**:

Algorithmic Formulation: Real-Time Estimated BOD Surrogate

$$BOD_{est}(t) = \alpha \cdot [\Delta DO_{deficit}(t) / \Delta t] + \beta \cdot Turbidity(t) + \gamma \cdot Conductivity(t) + \delta \cdot f(Temp)$$

Where DO_deficit measures deviation from 100% saturation at ambient temperature, Turbidity reflects suspended organic loading, and Conductivity scales with sewage concentration. The soft-sensor model is trained and calibrated against monthly certified laboratory BOD_5 incubation tests.

Parameters Relegated to Monthly Certified Lab Verification

PARAMETER	WHY EXCLUDED FROM IN-SITU LAKE SENSOR	PHASE 1 MEASUREMENT PROTOCOL	ML MODEL ROLE
Nitrate & Phosphate	ISE probes drift severely; wet-chemical analyzers require frequent reagent refills and technician servicing.	Certified lab testing on monthly grab samples collected at station.	Provides monthly training ground-truth to calibrate nutrient runoff models.
BOD_5	Requires 5-day laboratory incubation; physical sensor does not exist.	Standard 5-day dilution incubation at certified environmental laboratory.	Calibrates and validates the real-time soft-sensor BOD surrogate equation.
Fecal Coliforms	Automatic biosensors are high-cost, fragile, and research-stage only.	Monthly membrane filtration / CFU plating + weekly enzymatic field strips.	Ground-truth reference to train Pathogen Risk Index classifier.

6. Hardware, Firmware & Edge Upgrades: Resolving Bottlenecks

Nawdeep's review identified critical hardware constraints in earlier prototype designs. The engineering specification resolves each item as follows:

Microcontroller Upgrade: Migration to ESP32 — Fixed Single-ADC Bug

The Bottleneck: The ESP8266 / NodeMCU features only one analog input (A0). Wiring both analog pH and TDS probes simultaneously causes pin contention and crosstalk.
The Solution: Upgraded to the **ESP32 Dev Module** (dual-core 240 MHz, multiple 12-bit ADC channels). Probes connect via an external **ADS1115 16-bit I2C ADC** for superior noise rejection, or via industrial **RS-485 / Modbus RTU** over UART.

Lake Depth Measurement — Pressure vs. Ultrasonic

The JSN-SR04T ultrasonic module reaches a maximum of 500 cm (5 m) and suffers from acoustic wave scatter off windy water surfaces. For deep lake columns (>5m), an industrial **submersible hydrostatic pressure transducer** (0–10m range, 4–20mA output) is specified.

Air vs. Water Gas Sensors — MQ-136 / MQ-137 Context

MQ-series sensors (H_2S, NH_3) measure gas concentrations in air, not dissolved in water. They are explicitly excluded from submerged lake stations and reserved strictly for enclosed pump-house headspaces and sealed intake wet-wells.

Power Budget, Ingestion & Full-Stack Platform Architecture

SUBSYSTEM	COMPONENT & SPECIFICATION	OPERATIONAL FUNCTIONALITY
Solar & Battery	12V 500mA (10W) monocrystalline panel + 3.7V 18650 Li-ion pack (10,000 mAh) + MPPT charger.	Delivers **48 hours continuous autonomy** through extended monsoonal overcast periods. Edge deep-sleep duty cycling consumes <0.5 Wh/day.
Connectivity	SIM7600 4G LTE Cat-1 module with fall-back 2G/GSM; onboard LittleFS flash ring buffer.	Transmits encrypted telemetry every 10 minutes. Buffers up to 14 days of readings locally if cellular carrier drops, auto-resuming sync on reconnect.
Ingestion & Backend	MQTT over TLS (Mosquitto/EMQX) → **FastAPI (Python)** → **TimescaleDB**.	High-throughput time-series ingestion; TimescaleDB hypertables optimize continuous time-bucket aggregations and geospatial lake boundary queries.
Surveillance UI	**Next.js / React** + Tailwind CSS + Leaflet GIS Lake Map.	Interactive dashboard displaying live lake parameter curves, CPCB threshold warning lines, and one-click SMS alert dispatch to PHED engineers.

7. Machine Learning Pipeline & Strict Validation Protocol

The ML pipeline processes 10-minute telemetry data (~52,560 readings/sensor/year). To prevent data leakage and operational failure, models adhere to strict temporal validation:

1. **Quality Filter**
   Flags stuck values, out-of-range sensor spikes, and slow calibration drift; automatically excludes corrupt windows from training.
2. **Compensation**
   Applies real-time non-linear thermal compensation to pH, DO, and conductivity streams using digital temperature logs.
3. **Soft-Sensors**
   Estimates real-time BOD and nutrient proxies, dynamically adjusted against monthly certified laboratory incubation ground truth.
4. **Time Forecasting**
   Gradient boosted trees (XGBoost) forecast 48h dissolved oxygen collapse and algal bloom risk based on rain and weather APIs.

Validation Rule: No Random Train/Test Splits

In environmental time-series data, random k-fold splits leak future seasonal patterns into past training windows, artificially inflating model accuracy. Training and testing splits must strictly follow **Temporal Forward Chaining** (training on months 1–6, validating on month 7).

8. Maintenance Protocol, Calibration Cadence & Team Ownership

Sensors submerged continuously in biological lake water foul and drift. The deployment establishes an enforceable operational schedule:

CADENCE	OPERATIONAL ACTION	ASSIGNED ACCOUNTABILITY
Fortnightly (14 Days)	Physical visit to static station: inspect immersion assembly, clean optical DO lens, wipe turbidity window, remove algal biofilm from pH bulb.	**Field Maintenance Lead** (Designated Techno India NJR / Pilot Engineer).
Monthly (30 Days)	Two-point buffer calibration (pH 4.01 & 7.00); collect 1-litre grab sample for certified laboratory analysis (BOD_5, Nitrate, Coliform).	**Environmental Chemist / Lab Partner**.
Automated Continuous	Cloud-based algorithmic checks flag flatline sensor signals, sudden noise spikes, and abnormal baseline drift in the Next.js dashboard.	**Automated AarogyaJal Watchdog Daemon**.

Summary Synthesis: Ready for Fateh Sagar Deployment

By combining **Nawdeep's rigorous 6-sensor limnological baseline and hardware reality checks** with **AarogyaJal's ESP32 4G telemetry, 4-tier disease detection engine, and Next.js municipal alert platform**, the Phase 1 static station delivers a comprehensive, field-ready environmental and public health shield for Fateh Sagar Lake and Udaipur city.